

CORALL: A COLREGs-Guided Risk-Aware LLM for Decision-Making in Maritime Autonomous Surface Ships

Klinsmann Agyei, Wasif Naeem, Pouria Sarhadi

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The paper introduces a novel approach to utilising Large Language Models (LLMs) for real-time COLREGs-based decision-making in collision encounters. The COLREGs (collision regulations), published by the International Maritime Organisation (IMO), have served as baseline regulations for empirical maritime collision avoidance in open seas to ensure vehicle safety. As COLREGs were originally defined for human seafarers, translating them into computer language for autonomous operations has been extensively investigated in recent decades, attracting significant interest but lacking a universal solution. A key aspect is the need for explainable decision-making, either for autonomous operations or as captain assistance. This study leverages the power of LLMs to develop such a decision-making algorithm. The proposed COLREGs-guided Risk-Aware LLM (CORALL) algorithm consists of a high-level, risk-aware decision-maker and an execution layer that implements path tracking and collision avoidance in the presence of external disturbances. A framework for risk-aware, COLREGs-compliant LLM-based decision-making is devised, and its performance in online operation on a high-speed ship model is demonstrated. The LLM processes navigation outputs and risk indices, identifies the COLREGs encounter type, and makes decisions with accompanying explanations. The proposed solution is tested on all the 22 Imazu's benchmark problems, showcasing its effectiveness. This risk-aware AI solution, thought to be a first of its kind, opens the door to future research on advanced decision-making algorithms in maritime and similar operational environments.

Keywords

ai, col-regs, collision avoidance, computing and processing, decision making, llm, maritime autonomous surface ship, risk, robotics and control systems, transportation

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Index Terms—Maritime Autonomous Surface Ship, COLREGs, LLM, AI, Risk, Decision Making, Collision Avoidance.

I. INTRODUCTION

Maritime Autonomous Surface Ships (MASS) could significantly transform maritime operations, offering benefits such as enhanced safety, reduced costs, and lower carbon emissions through optimised route planning. Safe mission planning and decision-making are critical aspects for MASS, requiring global and local path planning, world modelling (including environmental conditions and spatial constraints), and consideration of vehicle dynamics and manoeuvrability limitations. MASS must also comply with the Convention on the International Regulations for Preventing Collisions at Sea (COLREGs) set by the IMO [1]. This involves safety-critical online risk assessment to detect collision risks and execute Collision Avoidance (COLAV) actions in line with these regulations. In 2022 alone, maritime accidents caused

losses exceeding \$1 billion, highlighting the need for safer navigation systems [2]. The 2021 Ever Given grounding in the Suez Canal caused weekly losses of \$10 billion, while the August 2024 collision between Chinese and Philippine vessels escalated tensions over COLREGs adherence [3]. Furthermore, a 10-year global forecast projects a significant growth in the autonomous ships market, increasing from \$6.1 billion to \$14 billion [4]. Autonomous ships, especially at the IMO’s full autonomy Level 4, must handle these complexities with greater reliability. However, the textual COLREGs, written for human interpretation, can be vague and challenging to apply in real-time, especially in multi-criteria decision-making situations that require rule interpretation across various scenarios. This complexity is further compounded by the need to develop decision-making capabilities that can translate qualitative COLREGs rules like ‘safe distance’, ‘good seamanship’, ‘in doubt’, ‘look-out’ and ‘ample time’, which require context-aware interpretation that varies with vessel characteristics and environmental conditions into actionable COLAV decisions. This challenge becomes particularly complex when making decisions in multi-vessel encounters, where MASS must assess multiple collision risks simultaneously while ensuring decisions remain compliant with COLREGs across various encounter types.

Recent advances in Machine Learning (ML) have shown promise in maritime decision-making [5], yet existing approaches often struggle to justify their COLAV decisions or maintain consistent COLREGs compliance. An early study, the COLREGs3 project [6], identified potential benefits of Large Language Models (LLMs) in this field, focusing on scene-based tests but lacking dynamic simulations. It was noted that this could lead to poor outcomes and rightly highlighted the need for re-training LLMs [6]. Similarly, research on advanced ship collision avoidance highlighted that introducing LLMs could represent a paradigm shift in ship decision-making [7]. Our initial work [8] demonstrated the potential of LLMs in maritime decision-making through validation across four single-vessel encounter types: head-on, overtaking, crossing from starboard, and crossing from port. However, real-world maritime environments frequently require decisions involving multiple vessels with varying risk levels, demanding robust integration of risk assessment with COLREGs interpretation.

The emergence of LLMs with advanced reasoning capabilities presents an exciting opportunity to address these multi-vessel challenges. Their ability to process natural language rules and generate contextual explanations, combined with real-time risk assessment, offers a promising solution for complex maritime navigation scenarios requiring sophisticated reasoning and clear decision justification.

Klinsmann Agyei and Pouria Sarhadi are with the School of Physics, Engineering and Computer Science, University of Hertfordshire, Hatfield, United Kingdom (k.agyei@herts.ac.uk, p.sarhadi@herts.ac.uk)

Wasif Naeem is with the School of Electronics, Electrical Engineering and Computer Science Queen’s University Belfast, Belfast, UK (w.naeem@qub.ac.uk)

This paper introduces CORALL, a novel framework that advances the application of LLMs in maritime autonomy by integrating risk-aware COLREGs-compliance decision-making with single and multi-vessel encounters. The system combines the interpretative capabilities of LLMs with quantitative risk metrics to generate explainable, regulation-compliant decisions for COLAV. Our key contributions include:

- Real-time interpretation of maritime situations with explicit COLREGs rule identification and risk assessment integration;
- Handling of multi-vessel encounters through risk-prioritised decision making;
- Clear, explainable reasoning for all decisions based on both regulatory requirements and quantitative risk metrics;
- Consistent performance evaluation across various encounter types, from single-vessel scenarios to complex multi-vessel interactions;

The remainder of this paper is organised as follows: Section II provides a review of related work across two key areas: COLREGs-based decision making, and applications of LLMs in autonomous decision systems. Section III presents our approach, beginning with CORALL's system architecture and core components, followed by the execution layer design, CORALL algorithm and decision-making framework, and implementation details that integrate these various components. Section IV presents extensive evaluation of results across the Imazu test scenarios, which serve as a widely recognised benchmark in maritime collision avoidance research. Finally, Section V concludes with a discussion of implications and future research directions.

II. RELATED STUDIES

A. COLREGs-Based Decision Making

Maritime navigation safety is fundamentally governed by the International Regulations for Preventing Collisions at Sea (COLREGs), established by the IMO [9]. As illustrated in Fig. 1, the application of COLREGs during a voyage follows a hierarchical decision-making process, beginning with basic safety assessments and progressing through various encounter scenarios. The relevant COLREGs rules for autonomous COLAV are as follows [1]:

1) **Rule 7: Risk of Collision:**

'Every vessel shall use all available means appropriate to the prevailing circumstances and conditions to determine if risk of collision exists. If there is any doubt, such risk shall be deemed to exist.'

2) **Rule 8: Action to Avoid Collision:**

'Any action taken to avoid collision shall be positive, made in ample time and with due regard to the observance of good seamanship.'

3) **Rule 13: Overtaking:**

'Any vessel overtaking any other shall keep out of the way of the vessel being overtaken.'

4) **Rule 14: Head-on Situation:**

'When two power-driven vessels are meeting on reciprocal or nearly reciprocal courses so as to involve risk of collision, each shall alter her course to starboard so that each shall pass on the port side of the other.'

5) **Rule 15: Crossing Situation:**

'When two power-driven vessels are crossing so as to involve risk of collision, the vessel which has the other on her starboard side shall keep out of the way and shall, if the circumstances of the case admit, avoid crossing ahead of the other vessel.'

6) **Rule 16: Action by Give-way Vessel:**

'Every vessel which is directed to keep out of the way of another vessel shall, so far as possible, take early and substantial action to keep well clear.'

7) **Rule 17: Action by Stand-on Vessel:**

'Where one of two vessels is to keep out of the way, the other shall keep her course and speed. The latter vessel may, however, take action to avoid collision by her manoeuvre alone, as soon as it becomes apparent to her that the vessel required to keep out of the way is not taking appropriate action.'

These rules form the foundation of maritime COLAV decision-making. However, their effective implementation in MASS presents significant challenges due to three key factors. First, COLREGs rely heavily on qualitative terms which require context-aware interpretation that varies with vessel characteristics and environmental conditions. Second, in multi-vessel scenarios, MASS must prioritise and coordinate responses to potentially conflicting COLREGs requirements, particularly challenging when different rules apply simultaneously to various encounter situations. Third, decisions must maintain temporal consistency to prevent oscillatory behaviour while remaining responsive to dynamic changes in the maritime environment.

Various approaches have attempted to address these challenges. Traditional geometric approaches established foundational frameworks, with Kuwata et al. [10] developing Velocity Obstacles (VO) methods for basic COLREGs-based decision-making. Campbell et al. [11] enhanced this approach by incorporating dynamic risk assessment through fundamental collision prediction metrics. However, these methods struggled to capture the nuanced reasoning required by COLREGs in complex scenarios. Rule-based systems emerged as a natural progression, attempting to directly encode COLREGs requirements [12]–[15]. Enhanced by fuzzy logic implementations [16], these systems better handle uncertainties but lack flexibility across diverse scenarios.

Recent ML approaches have shown promise, with deep reinforcement learning demonstrating capabilities in complex navigation decisions [17]–[19]. Probabilistic models and neural networks have enabled dynamic risk assessment [5], [20], though these AI-driven systems often operate as 'black boxes,' making it difficult to verify their reasoning or ensure COLREGs compliance.

Hybrid approaches have attempted to address these limitations by combining methodologies. Wang et al. [21] integrated geometric methods with ML to maintain explicit

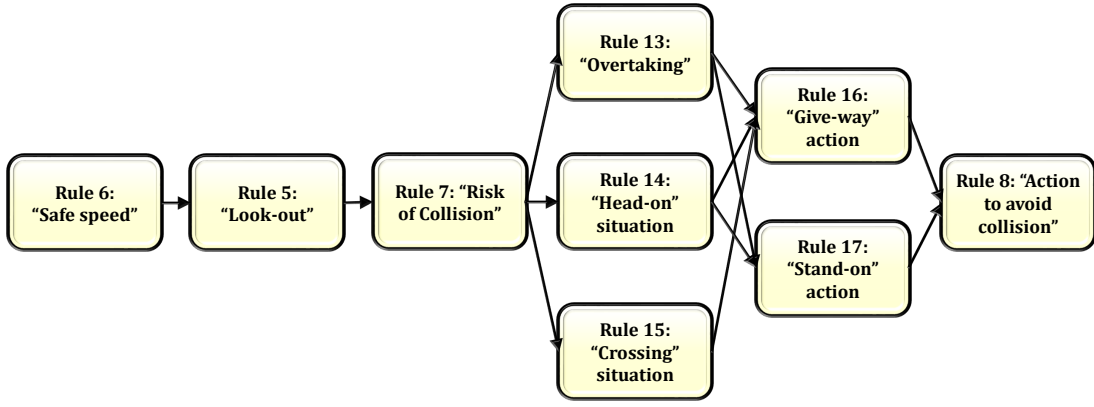


Fig. 1. Sequential flow of COLREGs, derived from IMO's textual rules [1]

COLREGs compliance. However, challenges remain in providing human-interpretable explanations and effectively handling complex multi-vessel encounters. Our work addresses these limitations through an LLM-based framework that bridges the gap between human-interpretable maritime regulations and autonomous control systems while maintaining acceptable risk assessment capabilities and the advantages of traditional methods in maritime navigation.

B. Application of LLMs in Autonomous Decision Making

Recently, there has been growing interest in exploiting LLMs for smart decision-making systems, with intensive research exploring their capabilities and limitations in addressing broader and more complex problems beyond Autonomous Vehicles (AVs) [22], [23]. Reference [23] highlights comparable traits of LLMs in solving financial and psychological problems, yet observes deficiencies in human-level decision-making for medical challenges. For instance, [24] reports that GPT-3.5's average performance in medical decision-making fell 10% short of the mean passing mark over the past two years. The research highlights that, to approach human expert-level performance, further development is required. Another study [25] reveals greater shortcomings when compared to top-tier and randomly selected doctors, though GPT's performance remains notable. As LLMs are still in their infancy, researchers anticipate better results in the future and are actively exploring improved learning and deployment strategies. Our study contributes to this by proposing a training process for real-time decision-making in COLREGs-compliant MASS navigation, a domain where LLM traits could be particularly beneficial given the written and subjective nature of these rules.

Exploiting existing LLMs in AVs presents two main challenges: i) enabling online and real-time decision-making, and ii) requiring commonly understood LLM re-training with an appropriate framework. In the self-driving car sector, companies like Wayve have demonstrated LLM integration to interpret complex traffic scenarios and generate natural language explanations for vehicle decisions [26]. Similarly, NVIDIA researchers have shown how LLMs can enhance perception and decision-making in AVs through their published work on language-conditioned driving policies [27].

These implementations have shown particular promise in handling edge cases and providing human-interpretable reasoning for vehicle actions. The adoption of LLMs extends beyond self-driving cars. In robotics, researchers have demonstrated successful applications in tasks ranging from manipulation planning to human-robot interaction. Notable work includes Code as Policies [28], which show how LLMs can effectively translate natural language commands into executable robot actions. The field of Unmanned Aerial Vehicles (UAVs) has similarly embraced LLM technology, with recent research demonstrating their effectiveness in mission planning, and regulatory compliance [29], [30]. Table I summarises the landscape of some notable LLM applications in autonomous systems, reflecting the considerable activities in this field. It provides insights into key attributes of each approach, including the application, input-output structure, and the model used. The survey reveals that GPT-3.5, GPT-4, and Llama are the predominant architectures, with GPT-4 being particularly favoured for its advanced reasoning capabilities. These implementations demonstrate diverse functionalities that parallel key challenges in maritime autonomy including:

- **Motion Planning:** LLMs process sensor data and generate trajectory plans, similar to our need for COLREGs-compliant path planning in maritime navigation;
- **Traffic Control:** Interpretation of complex traffic scenarios and rule compliance, directly analogous to multi-vessel encounter situations in maritime environments;
- **Policy Adaptation:** Dynamic adjustment of behaviour based on environmental conditions, reflecting the need for adaptive decision-making in varying maritime scenarios;
- **Natural Language Integration:** Processing rules and generating explanations, essential for interpreting qualitative COLREGs rules;
- **Safety Assessment:** Risk evaluation and safety-constraint verification, mirroring our requirements for maritime collision risk assessment;

These applications share a common thread of translating high-level rules into actionable decisions while maintaining explainability, a crucial requirement in maritime navigation where decisions must adhere to COLREGs while remaining interpretable to human operators. Building upon recent advances

TABLE I
COMPARISON BETWEEN LLM-BASED ALGORITHMS IN AV APPLICATIONS

Ref.	LLM Task	Input	Output	Model
Chen et al, 2024 [31]	Explainable driving (Driving with LLM)	Vectorised representation of objects in a scene and high level goals	Driving actions, explanations	GPT-3.5, LLaMa-7b-hf
Lin et al, 2024 [32]	Motion planner repair (Dr Planner)	Planner details, trajectories	Fixed planners	GPT-4-Turbo
Wang et al, 2024 [33]	Traffic signal control (LLM-Assisted light)	Traffic data	Signal control decisions	GPT-4
Li et al, 2024 [34]	Policy adaptation (LLaDA)	Traffic rules and scenarios	Adapted driving policies	GPT-4
Yildirim et al, 2024 [35]	Highway motion planning (HighwayLLM)	Current state and RL model output(meta level actions)	Waypoints and Trajectories	GPT-4
Zhong et al, 2024 [36]	Vision-based planning (UAVs with Dynamic Obstacle Trajectory Prediction)	Vision data, obstacle trajectories	Navigation commands	GPT-4
Ma et al, 2024 [37]	Code generation (Lampilot)	API documentation, human instructions, and driving context	Executable programs	Llama 2, PaLM 2, GPT-4
Fu et al, 2024 [38]	AV testing (LimSim++)	Visual Information, Scenario Cognition, Task Description	Driving decisions, reasoning	GPT-3.5, GPT-4, GPT-4V
Kong et al, 2024 [39]	Safe, compliant driving (Superalignment)	User requests, context, and safety rules	Safe, compliant, and potentially modified LLM responses	GPT-3.5, LLaMa2
Cui et al, 2024 [40]	Natural language driving (Drive as You Speak)	Natural language instructions and potentially contextual information	Driving commands	GPT-4
Fu et al, 2024 [41]	Human-like driving (Drive like a Human)	Scene context, past driving experience	Actions, explanations	GPT-3.5
Liang et al, 2023 [28]	Code generation (Code as Policies)	Natural language instructions, goals	Executable code	Codex code-davinci-002
Wang et al, 2023 [42]	Safety-focused planning	Scene info, actions	Safe plans	GPT-4
Wang et al, 2023 [43]	Instruction translation (Prompt a robot to walk)	Natural language prompts or instructions related to walking or locomotion	A sequence of actions or commands that the robot can execute	GPT-4
Wen et al, 2023 [44]	Knowledge-driven driving (DILU)	Sensor data, traffic information	Driving commands and policies	GPT-3.5, GPT-4
Sural et al, 2024 [45]	Context understanding (ContextVLM)	Visual, textual info	Scene context	ViLT, LLaVA
Tang et al, 2024 [46]	Scenario generation (LeGEND)	Scenario descriptions	Detailed scenarios	GPT-4
Cai et al, 2024 [47]	Regulation compliance (Driving with Regulation)	Traffic rules, action	Legality assessments	GPT-4o
Xu et al, 2024 [48]	Visual-based driving (DriveGPT4)	Images, instructions	Driving actions	GPT-4V
Deruyttere et al, 2022 [49]	Command translation (Talk2Car)	Driving commands	Physical trajectories	Sentence-BERT, RoBERT
Liang et al, 2023 [28]	Code generation (Code as Policies)	Natural language instructions, goals	Executable code	Codex code-davinci-002

in autonomous systems, we present CORALL, a novel framework that integrates LLM-based decision-making with real-time risk assessment for COLREGs-compliant autonomous navigation. Our approach uniquely combines the interpretative capabilities of LLMs with quantitative risk metrics to generate explainable, regulation-compliant decisions in maritime COLAV scenarios.

III. THE PROPOSED METHOD

This section presents CORALL, a novel framework that integrates LLMs with traditional maritime navigation systems to enable intelligent, risk-aware decision-making in autonomous surface vessels. CORALL processes real-time navigational data to generate COLREGs-compliant decisions. The system maintains state awareness across time steps, enabling consistent decision-making throughout vessel encounters while adapting to changing risk levels. The framework has been

evaluated across challenging scenarios, demonstrating its capability to handle various maritime situations from simple head-on encounters to complex multi-vessel scenarios. The following subsections detail the CORALL architecture, the execution layer, environment, CORALL algorithm and decision-making framework, and the implementation details.

A. CORALL System Architecture

The CORALL framework, illustrated in Fig. 5, comprises three distinct layers integrating navigation and perception, LLM-based decision processing, and execution control. This architecture ensures continuous risk assessment while maintaining regulatory compliance in autonomous maritime navigation. The Navigation and Perception Layer processes simulated navigational data that would typically be obtained from maritime sensors such as AIS, GNSS, IMU, RADAR, LIDAR, and Camera systems in real-world applications (not the target of this research). In our simulation environment, this layer

processes parameters including Distance at Closest Point of Approach (D_{CPA}), which represents the minimum predicted distance between vessels, Time to Closest Point of Approach (T_{CPA}), indicating the time until vessels reach D_{CPA} , Range (R), measuring the current distance between ships, and relative bearings (ψ_{rel}), forming a foundation for informed decision-making. The core intelligence resides in the LLM-Based Decision Processing Layer, which consists of three primary components:

- Data Preprocessing: Performs feature extraction and data normalisation, structuring the sensory inputs for LLM interpretation
- LLM Decision Core: Serves as the central reasoning unit, performing COLREGs rule identification, risk assessment, and situation explanation
- Decision Validation: Ensures generated decisions maintain both safety constraints and COLREGs compliance

The Execution Layer implements the validated decisions through three sequential components: Local Planner, Control, and Actuator. The Local Planner generates risk-aware trajectories based on the LLM's decisions, which are then translated into control commands and finally into specific actuator signals, as will be detailed in Section III.B. This hierarchical approach ensures implementation of COLAV manoeuvres while maintaining both safety margins and regulatory compliance. For multi-vessel scenarios, CORALL employs a risk-priority queue, processing targets based on their computed risk levels $Risk(t)_i$ and temporal constraints, ensuring highest-risk encounters receive immediate attention while maintaining overall situation awareness.

B. Execution layer and vehicle model

To understand the algorithm implementation process, it is crucial to explain the vehicle model and execution layer in Fig. 5. These can be replaced by any other functional model but were selected for their proven practical effectiveness.

C. Vehicle Dynamics

The MASS dynamics are modelled using a nonlinear model based on [12], [50]:

$$\begin{cases} \dot{X}(t) = u(t) \cos(\psi(t)) \\ \dot{Y}(t) = u(t) \sin(\psi(t)) \\ \dot{\psi}(t) = r(t) \\ \dot{r}(t) = -(1/T_\psi)r(t) + (K_\psi/T_\psi)\tau_c(t) \\ \dot{u}(t) = -(1/T_u)u(t) + (K_u/T_u)(\tau(t) - d(t)) \\ \dot{d}(t) = -(1/T_d)d(t) + \omega_d(t) \end{cases} \quad (1)$$

The state vector $[X, Y, \psi]$ represents the vessel's pose, where X and Y denote the longitudinal/lateral positions and ψ the heading angle. The yaw rate r follows Nomoto's first-order dynamics [50] with time constant T_ψ and gain K_ψ . The control input u_c is constrained within ± 20 units to reflect realistic actuator limitations. The vessel's surge speed u is governed by a first-order transfer function with parameters T_u and K_u . Environmental disturbances d are modelled via

a Markov process with Gaussian input ω_d and time constant T_d . This realistic model imitates most of the challenges in the dynamic behaviour of the MASS at this stage of design.

D. Local planning algorithm

The motion planning system integrates CORALL decisions with traditional navigation algorithms, based on [12], to execute COLREGs-compliant manoeuvres. As shown in Fig. 2, it is assumed that a global planner has generated the desired waypoints to track. The desired heading command ψ_d to follow the waypoints and avoid collisions combines three main components, as follows:

$$\psi_d(t) = \psi_{LOS}(t) + \psi_{CTE}(t) + \psi_{COLAV}(t) \quad (2)$$

These components are detailed as follows.

1) Line of sight tracking:

For basic path tracking, each waypoint is specified as $Wp(i) = [X_{wp(i)}, Y_{wp(i)}]$. As shown in Fig. 2, the tracking errors between current position and waypoint are defined as:

$$X_{ewp(i)}(t) = X_{wp(i)} - X(t) \quad (3)$$

$$Y_{ewp(i)}(t) = Y_{wp(i)} - Y(t) \quad (4)$$

The desired Line of Sight (LOS) angle to follow the waypoints is computed as:

$$\psi_{LOS}(t) = \text{atan2}(Y_{ewp(i)}(t), X_{ewp(i)}(t)) \quad (5)$$

2) Cross tracking error elimination:

To eliminate the cross tracking error (CTE), i.e., ϵ_{CTE} (shown in Fig. 2), the corresponding reference signal ψ_{CTE} is generated. To this end, the track length ($L_{TRK(i)}$) and track angle ($\psi_{TRK(i)}$) between successive waypoints are defined as:

$$L_{TRK(i)} = \sqrt{(X_{wp(i)} - X_{wp(i-1)})^2 + (Y_{wp(i)} - Y_{wp(i-1)})^2} \quad (6)$$

$$\psi_{TRK(i)} = \text{atan2}(Y_{wp(i)} - Y_{wp(i-1)}, X_{wp(i)} - X_{wp(i-1)}) \quad (7)$$

The along-track distance $S_i(t)$ is calculated as:

$$S_i(t) = \frac{((X_{wp(i)} - X_{wp(i-1)})X_{ewp(i)} + (Y_{wp(i)} - Y_{wp(i-1)})Y_{ewp(i)})}{L_{TRK(i)}} \quad (8)$$

Hence, the cross-track error is defined by:

$$\epsilon_{CTE}(t) = S_i(t) \tan(\delta_p(t)) \quad (9)$$

where $\delta_p(t) = \psi_{TRK(i)} - \psi_{LOS}$. Therefore, the CTE compensation term can be achieved using the following formula:

$$\psi_{CTE}(t) = -\text{atan}(\epsilon_{CTE}/\mu) \quad (10)$$

where μ is a tuning parameter [12].

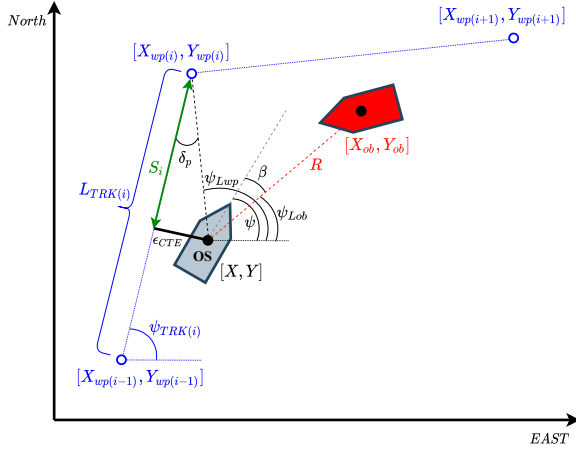


Fig. 2. Motion planning scene showing waypoints, track line, vessel position, cross-track error, and path angles.

3) CORALL-Guided Collision Avoidance:

The COLAV term incorporates CORALL's decisions and fuzzy weighting functions to enable dynamic obstacle prevention between Own Ship (OS) and Target Ship (TS). A collision encounter is illustrated in Fig. 3, which forms the basis for the derivation of the equations. As shown in Fig. 4, two primary weighting mechanisms are used:

- i) The distance-to-obstacle weighting uses a ZMF (Z-shaped Membership Function) as shown in Fig. 4 (left):

$$w_R = \begin{cases} 1, & R \leq a \\ 1 - 2\left(\frac{R-a}{b-a}\right)^2, & a < R \leq \frac{a+b}{2} \\ 2\left(\frac{R-b}{b-a}\right)^2, & \frac{a+b}{2} < R \leq b \\ 0, & R \geq b \end{cases} \quad (11)$$

- ii) The bearing-to-obstacle weighting uses a Gaussian function (Fig. 4, right):

$$w_\beta = \exp\left(-\frac{\beta^2}{2\sigma^2}\right) \quad (12)$$

where σ is the tuning parameter, β is the relative bearing, computed using the following equation:

$$\beta(t) = \psi(t) - \text{atan2}(Y_{ob}(t) - Y(t), X_{ob}(t) - X(t)) \quad (13)$$

The overall COLAV command for each obstacle integrates these weights with CORALL's decision:

$$\psi_{COLAV(i)} = K_{Dir} K_{COLAV} w_R w_\beta \quad (14)$$

The direction coefficient K_{Dir} is derived from the CORALL's interpreted navigation decisions, where a value of +1 corresponds to a starboard manoeuvre, -1 indicates a port turn, and 0 represents a stand-on situation, with the decision-making process detailed in the subsequent section. K_{COLAV} is a tuning parameter. For scenarios involving multiple targets, the total collision avoidance command is computed as:

$$\psi_{COLAV} = \sum_{i=1}^n \psi_{COLAV(i)} \quad (15)$$

This integration ensures that the LLM's high-level collision avoidance decisions are effectively translated into executable commands while to COLREGs compliance.

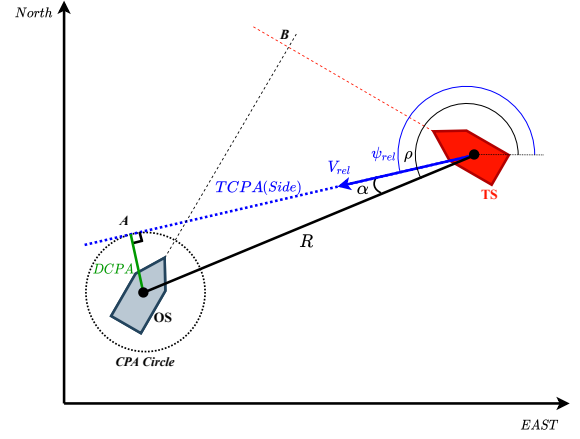


Fig. 3. A COLAV encounter illustrating OS, one TS, and CPA parameters.

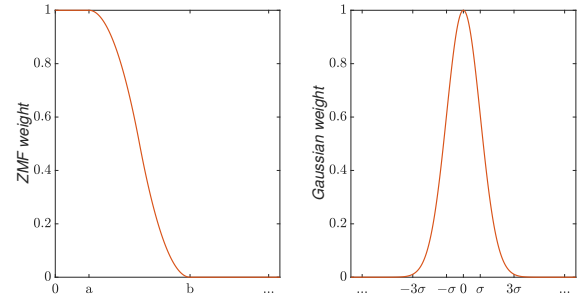


Fig. 4. Risk assessment functions: ZMF for distance-based risk and Gaussian for bearing-dependent risk.

E. The control algorithm

Planned manoeuvres are exerted through a PD controller, with gains adjustable based on vessel speed, expressed as follows:

$$\tau_c(t) = K_p(u)(\psi_d(t) - \psi(t)) - K_d(u)r(t) \quad (16)$$

Here, $K_p(u)$ and $K_d(u)$ are speed-dependent gains optimised for various operational conditions. Any advanced motion planning or control algorithm can be substituted as needed. Interested readers can refer to [12] for further details and the GitHub link for a Python implementation ¹.

F. CORALL Algorithm Formalisation

As formalised in Algorithm 1, CORALL processes navigation data for multiple vessels through a structured decision-making pipeline. For each target vessel i , the algorithm first constructs a state vector by calculating composite risk using fuzzy membership functions that combine D_{CPA} , T_{CPA} , and range parameters. This risk assessment informs the construction of the LLM input state vector $\mathcal{S}_{LLM}(t)$, which encapsulates all critical navigation parameters. The algorithm then implements target prioritisation based on calculated risk levels and temporal constraints, maintaining situational awareness

¹https://github.com/Psarhadi/Autonomous_ship_planning_collision_avoidance_control

Algorithm 1 CORALL: Risk-Aware LLM Decision Making for COLREGs Compliance

Require: Set of navigational data for each target $N_i = \{\psi_{rel}, R, D_{CPA}, T_{CPA}\}$ where $i \in \{1, \dots, n\}$ targets
Ensure: COLREGs-compliant outputs $\mathcal{L}(t) = \{\mathcal{R}(t), \mathcal{A}(t), \mathcal{E}(t)\}$
 /* State Vector Construction */
for each target i **do**
 Calculate composite risk $Risk_i$ using fuzzy membership functions
 $Risk_i \leftarrow (f(|D_{CPA}|, a_d, b_d) + f(|T_{CPA}|, a_t, b_t) + f(R(t), a_r, b_r))/3$
 Construct state vector $S_{LLM}(t) = [D_{CPA}(t), T_{CPA}(t), R(t), Risk(t), \psi(t)]$
end for
 /* Target Prioritisation */
 Identify critical target based on $Risk_i$ and temporal constraints
 Update state history $\mathcal{H}(t)$ and manoeuvre status $\mathcal{M}(t)$
 /* LLM Decision Generation */
 Construct prompt with $\mathcal{S}(t) = \{S_{LLM}(t), \mathcal{H}(t), \mathcal{M}(t)\}$
 Query LLM for structured output $\mathcal{L}(t)$
 Extract COLREGs rule $\mathcal{R}(t)$, action $\mathcal{A}(t)$, explanation $\mathcal{E}(t)$
 /* Direction Coefficient Extraction */
 Determine K_{Dir} from $\mathcal{A}(t)$ according to:

$$K_{Dir} = \begin{cases} +1, & \text{if starboard turn} \\ -1, & \text{if port turn} \\ 0, & \text{if stand-on} \end{cases}$$

return $\mathcal{L}(t) = \{\mathcal{R}(t), \mathcal{A}(t), \mathcal{E}(t)\}$

through state history $\mathcal{H}(t)$ and manoeuvre status $\mathcal{M}(t)$. These components are combined into a state representation $\mathcal{S}(t)$, which is used to construct the LLM prompt. The LLM processes this input to generate structured output $\mathcal{L}(t)$ comprising COLREGs rule identification and situation description $\mathcal{R}(t)$, action specification $\mathcal{A}(t)$, and decision explanation $\mathcal{E}(t)$. From the action component, the algorithm extracts a direction coefficient K_{Dir} that quantifies the required manoeuvre direction: +1 for starboard turn, -1 for port turn, and 0 for stand-on situations.

G. CORALL Decision-Making Framework

The design-time training process for the CORALL decision-making framework as illustrated in Fig. 6 comprises four interconnected components: prompt engineering with memory integration, CORALL decision maker, explainable action generation, and environmental feedback loop. This structure enables real-time, COLREGs-compliant navigation decisions while maintaining transparency and interoperability. The system's foundation lies in its prompt engineering framework, which incorporates both static task descriptions and dynamic memory components. The task description establishes the LLM's role as an autonomous ship's decision-making system, explicitly constraining its outputs to align with COLREGs requirements. This prompt template is augmented with a memory mechanism that maintains the temporal context of ongoing encounters, enabling the system to generate consistent decisions across multiple timesteps.

At the core of the architecture is the CORALL decision maker, implemented using LLMs. This component processes structured navigational data including relative bearing (ψ_{rel}), range (R), T_{CPA} and D_{CPA} . The decision maker analyzes these parameters along with state history and manoeuvre status within the context of COLREGs rules to evaluate the current

situation and determine appropriate actions. The system generates explainable actions through a structured output format that includes:

- COLREGs rule identification and situation description
- Specific action recommendation (Stand-on/Give-way decisions)
- Manoeuvre specification when required (e.g. starboard turn)
- Risk assessment and justification

An important feature of the architecture is its environmental feedback loop, which connects the control system's execution of manoeuvres back to the decision-making process. This feedback mechanism enables the LLM to adapt its decisions based on the effectiveness of previous actions and evolving navigational conditions.

The integration of these components creates a closed-loop decision-making system that maintains COLREGs compliance while providing human-interpretable explanations for all navigation decisions. This transparency is particularly valuable for the verification of autonomous maritime systems, where understanding the reasoning behind navigational choices is crucial for safety and regulatory compliance. The implementation of CORALL, including the complete simulation environment and test scenarios, is available in our GitHub repository².

H. Implementation

The implementation of CORALL combines the previously described components into an integrated system, bringing together real-time risk assessment with LLM-based decision-making for autonomous COLREGs-compliant navigation. While other LLMs such as LLaMA 3.1 and PaLM-2 were considered, GPT-4 was selected based on initial testing of reasoning capabilities and decision consistency, though a comprehensive comparative analysis remains for future investigation. This implementation unifies the architectural elements into a cohesive framework that extends beyond traditional autonomous navigation systems by incorporating explainable decision-making while maintaining real-time performance. The algorithm underwent supervised training, with errors or improvements reported to CORALL being recorded in the LLM's memory. The run-time CORALL algorithm employs the same prompt engineering, interacting solely with the environment to generate safe actions, as illustrated in Fig. 6. Some of these prompts are revealed in Section IV.

1) Online risk-assessment through CPA Calculations

In maritime collision avoidance decision making, the Closest Point of Approach (CPA) methodology stands as the most widely accepted approach for risk analysis [12]. Our implementation adopts and extends this fundamental technique, incorporating real-time risk assessment into the LLM decision-making process.

As illustrated in Fig. 3, the CPA calculation considers a Target Ship (TS) approaching Own Ship (OS), where the CPA point represents the tangent to a circle centered at the OS along the relative velocity vector V_{rel} . The critical parameters D_{CPA} and T_{CPA} are computed using:

²<https://github.com/Klins101/CORALL.git>

$$D_{CPA}(t) = R(t) \sin(\alpha(t)) \quad (17)$$

$$T_{CPA}(t) = R(t) \cos(\alpha(t)) / V_{rel}(t) \quad (18)$$

The relative velocity vector components are determined by:

$$V_{rel}(t) = \sqrt{(V_{xob}(t) - V_x(t))^2 + (V_{yob}(t) - V_y(t))^2} \quad (19)$$

$$\psi_{vrel}(t) = \text{atan2}(V_{yob}(t) - V_y(t), V_{xob}(t) - V_x(t)) \quad (20)$$

where $[V_x(t), V_y(t)]$ and $[V_{xob}(t), V_{yob}(t)]$ represent the velocity vectors of OS and TS respectively.

2) Risk Evaluation

Our system implements a composite risk assessment using Z-shaped Membership Functions (ZMF) to normalise and combine multiple risk factors:

$$Risk(t) = \frac{|f_D(D_{CPA})| + |f_T(T_{CPA})| + f_R(R(t))}{3} \quad (21)$$

where:

$$\begin{aligned} f_D(x) &= f(x, a_{D_{CPA}}, b_{D_{CPA}}) \\ f_T(x) &= f(x, a_{T_{CPA}}, b_{T_{CPA}}) \\ f_R(x) &= f(x, a_R, b_R) \end{aligned}$$

where $f(x, a_{D_{CPA}}, b_{D_{CPA}})$ represents the ZMF with thresholds $a_{D_{CPA}}$ and $b_{D_{CPA}}$. For D_{CPA} , $a_{D_{CPA}} = 0.2$ nmi and $b_{D_{CPA}} = 0.5$ nmi; for T_{CPA} , $a_{T_{CPA}} = 1$ min and $b_{T_{CPA}} = 4$ min; and for range, $a_R = 0.1$ nmi and $b_R = 0.3$ nmi. This implementation offers several advantages:

- Normalisation of diverse parameters (distance, time) to a $[0,1]$ scale
- Smooth transition between risk levels through ZMF
- Configurable risk thresholds for different operational contexts
- Unified risk representation for LLM decision-making

This risk assessment methodology directly informs the LLM's decision-making process, enabling quantitative risk evaluation while maintaining interpretable outputs for collision avoidance decisions.

3) Decision Integration

State Management: The integration between CORALL's decision-making and motion planning is implemented through a state management system, as detailed in Algorithm 1. Real-time navigation parameters and risk metrics are continuously processed and structured for CORALL consumption. The system preserves temporal consistency by monitoring both current navigational states and historical encounter data, enabling the LLM to generate contextually aware decisions. This state management ensures seamless transitions between decisions while preventing oscillatory behaviour in complex scenarios.

CORALL-Motion Planner Interface The interface facilitates bidirectional communication between CORALL's high-level decision-making and the motion planning system. LLM decisions are translated into executable commands through the direction coefficient K_{Dir} , which quantifies manoeuvre intentions into specific control inputs as shown in equation 16. The system generates risk-aware trajectories that balance COLREGs compliance with safety constraints, continuously monitoring execution performance through a feedback loop. This monitoring enables dynamic adaptation of decisions based on the effectiveness of ongoing manoeuvres and evolving navigational conditions. The integrated framework maintains decision consistency across multiple time steps while adapting to changing risk levels. This tight coupling between decision-making and execution layers enables the system to handle complex multi-vessel encounters while maintaining clear reasoning chains for all navigation decisions.

IV. SIMULATION RESULTS

The simulation environment includes the system architecture and algorithms detailed in Section III, with the execution layer operating at 100Hz frequency for motion control and trajectory updates. The LLM-based decision-making runs at 1Hz frequency, providing sufficient responsiveness while ensuring computational feasibility for real-time implementation. The system integrates COLREGs-compliant decision-making with traditional motion planning through the architecture shown in Fig. 5.

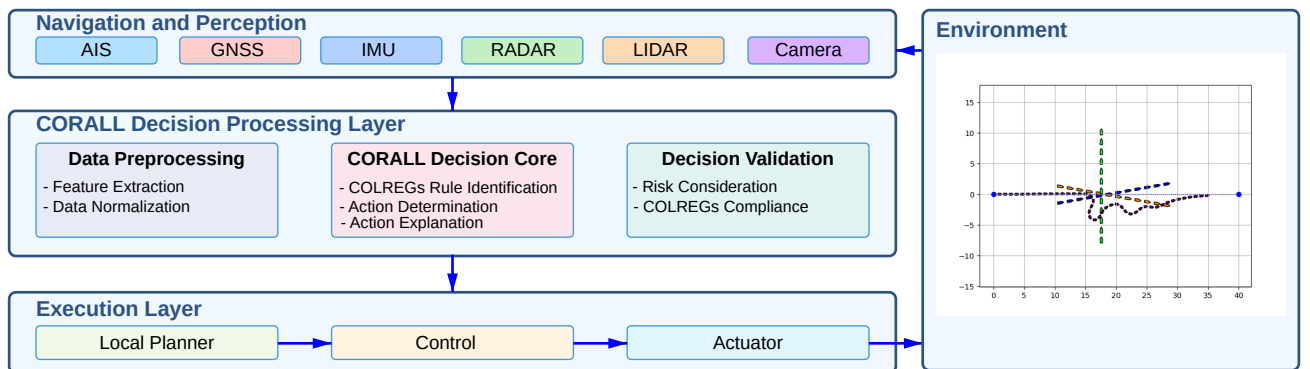


Fig. 5. Architecture of the proposed CORALL

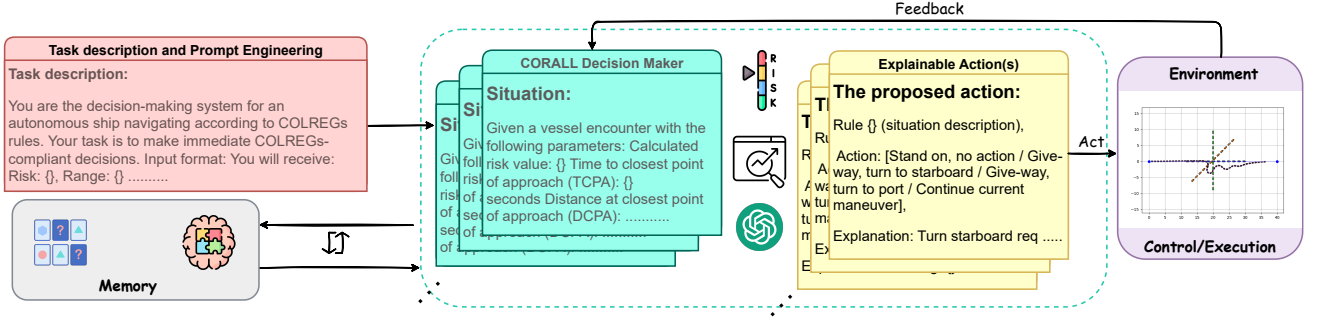


Fig. 6. The design-time architecture of CORALL

TABLE II
INITIAL PARAMETERS FOR IMAZU TEST SCENARIOS

Case	Scenario Type	Target 1	Target 2	Target 3
1	Head-on	(6, 0), 180°	—	—
2	Crossing	(5, -2), 90°	—	—
3	Overtaking	(3, 0), 0°	—	—
4	Crossing	(3.44, 1.5), 295°	—	—
5	Crossing + Head-on	(5, -2.01), 90°	(6.95, 0), 180°	—
6	Multiple Crossing	(3.4, -1.35), 45°	(3, -0.35), 10°	—
7	Overtaking + Crossing	(3, 0), 0°	(3.4, -1.35), 45°	—
8	Crossing + Head-on	(5, -2.01), 90°	(7, 0), 180°	—
9	Multiple Crossing	(3.4, -1.34), 45°	(5, -2.0), 90°	—
10	Multiple Crossing	(3, 0.35), 350°	(4.4, -1.78), 90°	—
11	Multiple Crossing	(5, 2.0), -90°	(3.4, -1.35), 45°	—
12	Head-on + Crossing + Crossing	(7, 0), 180°	(3, 0.35), -10°	(3.44, -1.35), 45°
13	Head-on + Crossing + Crossing	(6, 0), 180°	(3, 0.35), 350°	(3.4, 1.47), 295°
14	Triple Crossing	(3.4, -1.35), 45°	(3, -0.35), 10°	(5, -2.0), 90°
15	Overtaking + Crossing + Crossing	(3, 0), 0°	(3.4, -1.35), 45°	(5, -2.0), 90°
16	Triple Crossing	(3.4, 1.35), -45°	(5, 2.0), -90°	(5, -2.0), 90°
17	Overtaking + Crossing + Crossing	(3, 0), 0°	(3, 0.35), -10°	(3.4, -1.34), 45°
18	Triple Crossing	(3.3, -0.4), 10°	(3.4, -1.35), 45°	(6.5, -1.47), 135°
19	Triple Crossing	(3, -0.35), 10°	(3, 0.35), -10°	(6.5, -1.42), 135°
20	Overtaking + Crossing + Crossing	(3, 0), 0°	(3, -0.35), 10°	(4.4, -1.78), 90°
21	Triple Crossing	(2.7, -0.35), 10°	(2.7, 0.32), -10°	(4.4, -1.78), 90°
22	Overtaking + Crossing + Crossing	(3, 0), 0°	(3.94, -1.73), 45°	(5, -2.0), 90°

The performance evaluation is conducted using the standardised Imazu test scenarios [51], which are now widely accepted as benchmarks in maritime collision avoidance research [5], [52], [53]. Our system was tested with the own ship starting at $(X(t_0), Y(t_0)) = (0, 0)$ nautical miles, heading $\psi(t_0) = 0$ degrees with velocity of 25 knots, and waypoint at $(X_{wp}(0), Y_{wp}(0)) = [90, 0]$ nautical miles. Target vessels move at 10 knots, with initial coordinates adapted from [15], with modifications to set initial D_{CPA} values near zero, thereby increasing collision risk (see Table II).

The trajectories across all 22 test scenarios are shown in Fig. 7, where the own ship (OS) trajectory appears in black, with target ships (TS) in blue (TS1), orange (TS2), and green (TS3) based on vessel count. Fig. 8 presents the time histories of key safety parameters (D_{CPA} , T_{CPA} , Range, and Risk), demonstrating successful collision avoidance across scenarios ranging from basic head-on encounters to complex

multi-vessel interactions. The collision risk parameters were monitored within specific ranges: D_{CPA} within $[0.2, 0.5]$ nautical miles, T_{CPA} between $[2, 4]$ minutes, and obstacle distances varying from $[0.1, 0.3]$ nautical miles. Notably, the small variations in the D_{CPA} demonstrate the algorithm's robustness in handling intentionally introduced disturbances.

For a more detailed analysis, we also present three representative scenes involving the generated explanations demonstrating the performance of the CORALL.

A. Case 1: Head-on Encounter

In this fundamental test of COLREGs Rule 14 application, CORALL demonstrated successful decision-making throughout the encounter. The parameter evolution plots in Fig. 8 reveal key aspects of the system's performance. The D_{CPA} plot shows relatively low values initially, followed by a significant increase at around $t = 500$ s as a result of the collision

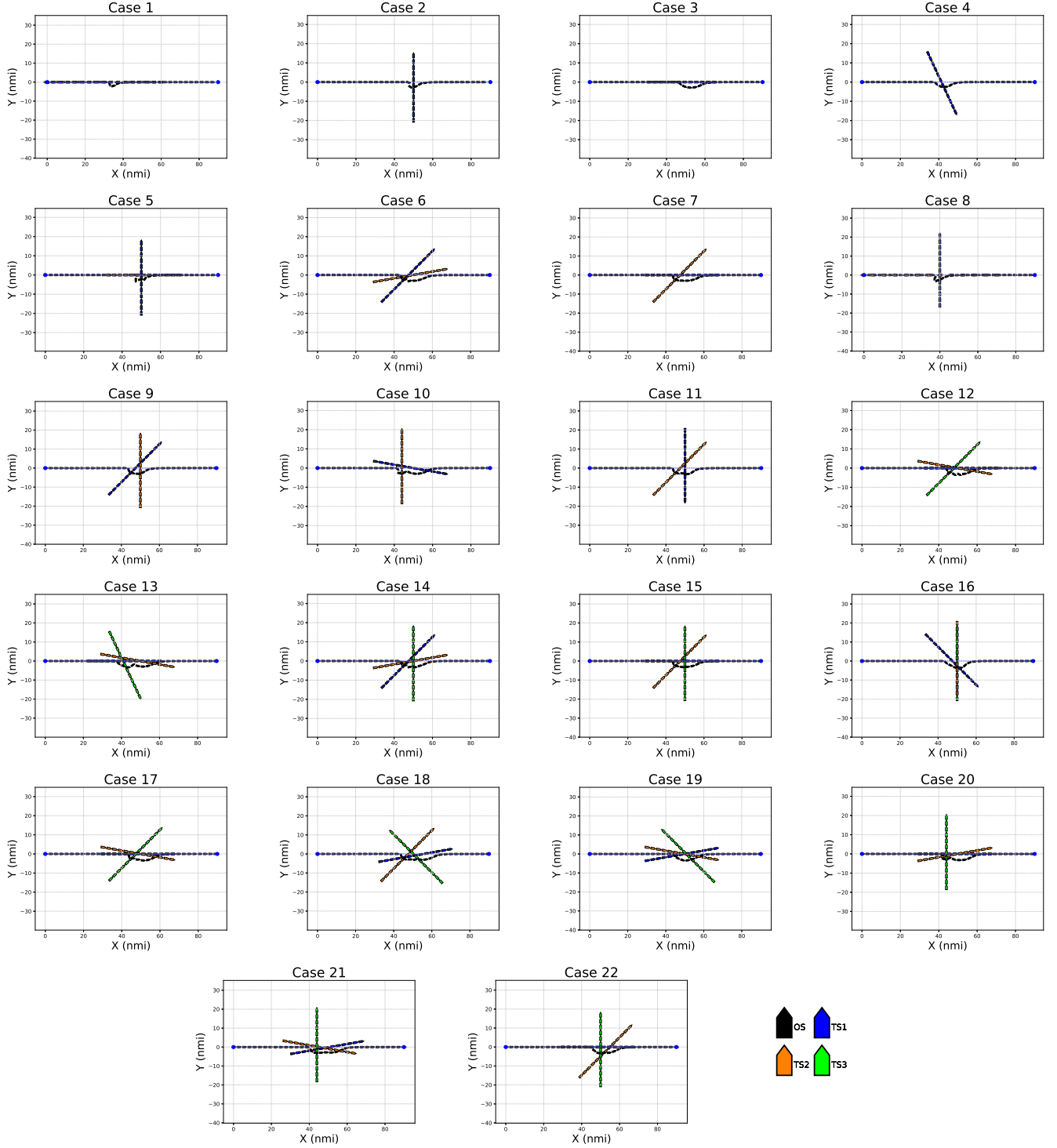


Fig. 7. Trajectories of the own ship and dynamic obstacles for all 22 Imazu cases

avoidance manoeuvre. The Risk profile shows moderate levels (around 0.6) before the manoeuvre, with a spike during closest approach, followed by a reduction to safer levels after successful avoidance action. As shown in the decision visualisation in Fig.9 (input-output prompts), at a critical point in the encounter, CORALL correctly identified a head-on situation based on the relative bearing of 180.0° with high risk (0.71). The system generated a "give-way, turn starboard" decision, adhering to COLREGs Rule 14 requirements. The trajectory plot in Fig.9 (left) demonstrates the successful execution of

this decision, as evidenced by the black vessel's (representing own ship) clear starboard turn away from its original course. This manoeuvre effectively prevented a collision with the target vessel, whilst maintaining COLREGs compliance, as confirmed by the subsequent improvement in risk metrics shown in Fig. 8 (case 1).

B. Case 7: Overtaking with Crossing

In Case 7, CORALL processes a two-vessel encounter scenario as shown in Fig.7(case 7). At the captured instance



Fig. 8. Plot of parameter evolution in all 22 Imazu cases.

shown in Fig.10 (left), the system evaluates TS1 (blue) ahead and TS2 (orange) crossing. Although both vessels present moderate risk levels, CORALL identifies the critical nature of TS2's close proximity (0.27 nmi Range, 0.28 nmi DCPA), generating a "give-way, turn starboard" decision in compliance with Rule 15. The trajectory plot in Fig.10 demonstrates successful execution of this decision, with the black vessel (OS) clearly executing the instructed starboard turn to avoid collision. The parameter evolution in Fig. 8 (case 7) shows this decision-making process. The $DCPA$ plot shows distinct

patterns for both vessels, with notable improvements following the avoidance manoeuvre between $t = 500s$ and $t = 1000s$. The Risk profiles demonstrate CORALL's ability to maintain separate risk assessments for each vessel while executing appropriate avoidance actions. The TCPA trends and Range plots confirm the successful management of both encounters, with safe distances maintained throughout the scenario.

C. Case 22: Complex Multi-vessel Encounter

The most complex scenario (Case 22) demonstrates CORALL's capability in handling three-vessel encounters, as shown in Fig.11 (left). At the captured instance, with three vessels in proximity (blue TS1, orange TS2, and green TS3), CORALL identifies a critical situation (input-output prompts of Fig.11) where TS3 presents moderate risk (0.67) at close range (0.35 nmi Range, 0.145 nmi DCPA). The trajectory plot in Fig.11 (left) shows the black vessel (OS) executing the recommended starboard turn, successfully avoiding collision while maintaining COLREGs compliance. The parameter evolution in Fig. 8 (case 22) shows this multi-vessel management strategy. The D_{CPA} plots show coordinated patterns for all three vessels, with successful avoidance manoeuvres reflected in the improvements between $t = 500$ s and $t = 1000$ s. The Risk profiles, maintained below 0.7 throughout the encounter, demonstrate CORALL's ability to handle multiple simultaneous threats while executing appropriate avoidance actions. The TCPA trends confirm effective time management across all three vessel interactions.

These results demonstrate the effectiveness of CORALL approach across the full spectrum of maritime encounter scenarios, from simple single-vessel situations to complex multi-target encounters. The system's ability to maintain COLREGs compliance while providing clear, explainable decisions demonstrates its potential for practical maritime applications.

V. CONCLUSION

This paper presented CORALL, a novel framework that integrates LLM with risk-aware decision-making for autonomous maritime navigation. Our approach successfully addresses three key challenges in autonomous vessel operation: interpreting qualitative COLREGs rules, maintaining consistent risk awareness, and generating explainable navigation decisions. Through extensive evaluation across 22 standardised Imazu test scenarios, CORALL demonstrated acceptable performance in:

- Real-time COLREGs interpretation and compliance
- Risk-aware decision-making in complex multi-vessel encounters
- Clear explanation generation for all navigation decisions
- Consistent behaviour across varying maritime situations

The integration of CORALL reasoning with traditional risk assessment metrics provides several key advantages:

- Direct translation of qualitative maritime regulations into quantitative actions
- Dynamic adaptation to varying risk levels while maintaining regulatory compliance

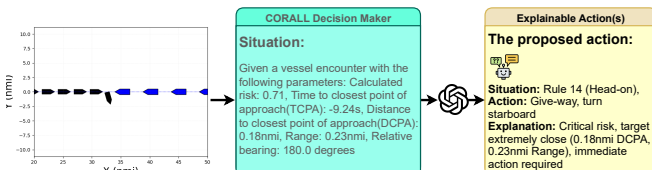


Fig. 9. LLM decision-making process for head-on encounter (Case 1)

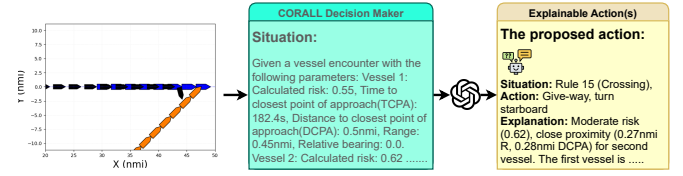


Fig. 10. LLM decision-making process for overtaking with crossing (Case 7)

- Human-interpretable explanations for all decisions
- Scalable handling of multiple vessel encounters

A. Limitations

While CORALL demonstrates promising results across whole Imazu test scenarios and beyond [8], certain limitations or uncertainties should be acknowledged. The current implementation has been evaluated in simulation environments with idealised sensor data, which may not fully reflect real-world performance. However, a realistic control model, incorporating actuation limitations and environmental disturbances, has been considered. Additionally, while the LLM provides explainable decisions, its computational requirements may pose challenges for real-time implementation in resource-constrained maritime systems. Although CORALL demonstrates acceptable performance in standardised test scenarios, it may face challenges in more complex situations exceeding the complexity level of the Imazu test cases, particularly in scenarios with intricate environmental interactions. This limitation is common among most algorithms proposed in the literature and should be addressed in future work. The system's performance in such edge cases needs further investigation. Nevertheless, this paper asserts the feasibility of the proposed approach as an essential step forward.

B. Future Work

Future research can be pursued in several directions. Regardless of the results of this study, whilst Imazu's test cases are widely accepted within the maritime COLAV community, a more rigorous approach, such as Monte Carlo simulations and a more comprehensive testing framework with diverse scenarios, should be employed to ensure seamless and robust performance testing. This can also involve formal verification approaches. Demonstrating the real-time implementability of the algorithm is another promising avenue, which our early studies indicate is feasible and will be proved in the future.

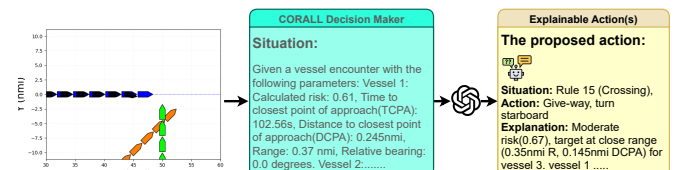


Fig. 11. LLM decision-making process for overtaking with multi-vessel crossing (Case 22)

The application of Small Language Models (SLMs) or customised models to problems such as COLREGs could also deliver impact by designing dedicated models with improved awareness and reduced computational burden. The insights provided by this research can support the development of a better framework for explainable and intelligent decision-making in MASS applications.

REFERENCES

- [1] International Maritime Organization (IMO), *Convention on the International Regulations for Preventing Collisions at Sea, 1972 (COLREGs)*. London: IMO, 1972.
- [2] A. McMillan, "Shipping losses drop to all-time low – allianz commercial," <https://supplychaindigital.com/supply-chain-risk-management/allianz-commercial-safety-shipping-review-2024>, 2024, accessed: 28th September 2024.
- [3] C. Clark. (2024, Aug.) After ship collisions, experts say china testing philippine resolve, us ties. Accessed: 15-Sep-2024. [Online]. Available: <https://breakingdefense.com/2024/08/after-ship-collisions-experts-say-china-testing-philippine-resolve-us-ties/>
- [4] S. Gujar and D. Vishwakarma, "Autonomous ships market size - by ship type, by offering, by propulsion, by autonomy, by end-user & forecast, 2024-2032," <https://www.gminsights.com/industry-analysis/autonomous-ships-market>, Aug. 2024.
- [5] P. Sarhadi, W. Naeem, and N. Athanasopoulos, "A survey of recent machine learning solutions for ship collision avoidance and mission planning," *IFAC-PapersOnLine*, vol. 55, no. 31, pp. 257–268, 2022.
- [6] L. Sanchez-Heres, R. Weber, F. Ahlgren, F. Olsson, and O. Lundström, "Colreg 3 - exploring the potential of large language models in marine navigation systems," <https://research.chalmers.se/en/publication/540274>, February 2024.
- [7] Y. Wang and Y. Zhao, "Multiple ships cooperative navigation and collision avoidance using multi-agent reinforcement learning with communication," *Ocean Engineering*, vol. 320, p. 120244, 2025.
- [8] K. Agyei, P. Sarhadi, and W. Naeem, "Large language model-based decision-making for colregs and the control of autonomous surface vehicles," *arXiv preprint arXiv:2411.16587*, 2024.
- [9] J.-K. Kim and D.-J. Park, "Understanding of sailing rule based on colregs: Comparison of navigator survey and automated collision-avoidance algorithm," *Marine Policy*, vol. 159, p. 105894, 2024.
- [10] F. Zhang, Y. Liu, L. Du, F. Goerlandt, Z. Sui, and Y. Wen, "A rule-based maritime traffic situation complex network approach for enhancing situation awareness of vessel traffic service operators," *Ocean Engineering*, vol. 284, p. 115203, 2023.
- [11] Y. Huang, L. Chen, P. Chen, R. R. Negenborn, and P. Van Gelder, "Ship collision avoidance methods: State-of-the-art," *Safety science*, vol. 121, pp. 451–473, 2020.
- [12] P. Sarhadi, W. Naeem, and N. Athanasopoulos, "An integrated risk assessment and collision avoidance methodology for an autonomous catamaran with fuzzy weighting functions," in *2022 UKACC 13th International Conference on Control (CONTROL)*. IEEE, 2022, pp. 228–234.
- [13] Y. A. Ahmed, M. A. Hannan, M. Y. Oraby, and A. Maimun, "Colregs compliant fuzzy-based collision avoidance system for multiple ship encounters," *Journal of Marine Science and Engineering*, vol. 9, no. 8, p. 790, 2021.
- [14] M. Lysak, G. Silva, C. Almeida, A. Martins, and J. Almeida, "Colreg compliant collision avoidance system for an unmanned surface vehicle," in *OCEANS 2024-Singapore*. IEEE, 2024, pp. 01–06.
- [15] B. Singh, K. A. Dastgerdi, N. Athanasopoulos, W. Naeem, and B. Lecallard, "Provably safe finite-time guidance for marine vehicles," *arXiv preprint arXiv:2402.06291*, 2024.
- [16] M. Aydin, E. Akyuz, O. Turan, and O. Arslan, "Validation of risk analysis for ship collision in narrow waters by using fuzzy bayesian networks approach," *Ocean Engineering*, vol. 231, p. 108973, 2021.
- [17] L. Zhao and M.-I. Roh, "Colregs-compliant multiship collision avoidance based on deep reinforcement learning," *Ocean Engineering*, vol. 191, p. 106436, 2019.
- [18] C. Wang, X. Zhang, H. Gao, M. Bashir, H. Li, and Z. Yang, "Colregs-constrained safe reinforcement learning for realising mass's risk-informed collision avoidance decision making," *Knowledge-Based Systems*, vol. 300, p. 112205, 2024.
- [19] X. Xu, Y. Lu, X. Liu, and W. Zhang, "Intelligent collision avoidance algorithms for usvs via deep reinforcement learning under colregs," *Ocean Engineering*, vol. 217, p. 107704, 2020.
- [20] H. Fan, H. Jia, X. He, and J. Lyu, "Navigating uncertainty: A dynamic bayesian network-based risk assessment framework for maritime trade routes," *Reliability Engineering & System Safety*, vol. 250, p. 110311, 2024.
- [21] W. Wang, L. Huang, K. Liu, X. Wu, and J. Wang, "A colregs-compliant collision avoidance decision approach based on deep reinforcement learning," *Journal of Marine Science and Engineering*, vol. 10, no. 7, p. 944, 2022.
- [22] Y. Huang, L. Sun, H. Wang, S. Wu, Q. Zhang, Y. Li, C. Gao, Y. Huang, W. Lyu, Y. Zhang *et al.*, "Trustllm: Trustworthiness in large language models," *arXiv preprint arXiv:2401.05561*, 2024.
- [23] B. Guo, X. Zhang, Z. Wang, M. Jiang, J. Nie, Y. Ding, J. Yue, and Y. Wu, "How close is chatgpt to human experts? comparison corpus, evaluation, and detection," *arXiv preprint arXiv:2301.07597*, 2023.
- [24] A. J. Thirunavukarasu, R. Hassan, S. Mahmood, R. Sanghera, K. Barzangi, M. El Mukashfi, and S. Shah, "Trialling a large language model (chatgpt) in general practice with the applied knowledge test: observational study demonstrating opportunities and limitations in primary care," *JMIR Medical Education*, vol. 9, no. 1, p. e46599, 2023.
- [25] R. Arvidsson, R. Gunnarsson, A. Entezarjou, D. Sundemo, and C. Wikberg, "Chatgpt (gpt-4) versus doctors on complex cases of the swedish family medicine specialist examination: an observational comparative study," *BMJ open*, vol. 14, no. 12, p. e086148, 2024.
- [26] R. Chen, W. Song, W. Zu, Z. Dong, Z. Guo, F. Sun, Z. Tian, and J. Wang, "An llm-driven framework for multiple-vehicle dispatching and navigation in smart city landscapes," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 2147–2153.
- [27] J. Roh, C. Paxton, A. Pronobis, A. Farhadi, and D. Fox, "Conditional driving from natural language instructions," in *Conference on Robot Learning*. PMLR, 2020, pp. 540–551.
- [28] J. Liang, W. Huang, F. Xia, P. Xu, K. Hausman, B. Ichter, P. Florence, and A. Zeng, "Code as policies: Language model programs for embodied control," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2023, pp. 9493–9500.
- [29] S. Javadi, H. Fahim, B. He, and N. Saeed, "Large language models for uavs: Current state and pathways to the future," *IEEE Open Journal of Vehicular Technology*, 2024.
- [30] J. Ioannidis, J. Harper, M. S. Quah, and D. Hunter, "Gracenote. ai: Legal generative ai for regulatory compliance," in *Proceedings of the Third International Workshop on Artificial Intelligence and Intelligent Assistance for Legal Professionals in the Digital Workplace (LegalAIIA 2023)*, 2023.
- [31] L. Chen, O. Sinavski, J. Hünermann, A. Karnsund, A. J. Willmott, D. Birch, D. Maund, and J. Shotton, "Driving with llms: Fusing object-level vector modality for explainable autonomous driving," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 14 093–14 100.
- [32] Y. Lin, C. Li, M. Ding, M. Tomizuka, W. Zhan, and M. Althoff, "Drplanner: Diagnosis and repair of motion planners for automated vehicles using large language models," *IEEE Robotics and Automation Letters*, 2024.
- [33] M. Wang, A. Pang, Y. Kan, M.-O. Pun, C. S. Chen, and B. Huang, "Llm-assisted light: Leveraging large language model capabilities for human-mimetic traffic signal control in complex urban environments," *arXiv preprint arXiv:2403.08337*, 2024.
- [34] B. Li, Y. Wang, J. Mao, B. Ivanovic, S. Veer, K. Leung, and M. Pavone, "Driving everywhere with large language model policy adaptation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 14 948–14 957.
- [35] M. Yildirim, B. Dagda, and S. Fallah, "Highwayllm: Decision-making and navigation in highway driving with rl-informed language model," *arXiv preprint arXiv:2405.13547*, 2024.
- [36] J. Zhong, M. Li, Y. Chen, Z. Wei, F. Yang, and H. Shen, "A safer vision-based autonomous planning system for quadrotor uavs with dynamic obstacle trajectory prediction and its application with llms," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2024, pp. 920–929.
- [37] Y. Ma, C. Cui, X. Cao, W. Ye, P. Liu, J. Lu, A. Abdelraouf, R. Gupta, K. Han, A. Bera *et al.*, "Lampilot: An open benchmark dataset for autonomous driving with language model programs," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 15 141–15 151.

- [38] D. Fu, W. Lei, L. Wen, P. Cai, S. Mao, M. Dou, B. Shi, and Y. Qiao, "Limsim++: A closed-loop platform for deploying multimodal llms in autonomous driving," *arXiv preprint arXiv:2402.01246*, 2024.
- [39] X. Kong, T. Braunl, M. Fahmi, and Y. Wang, "A superalignment framework in autonomous driving with large language models," *arXiv preprint arXiv:2406.05651*, 2024.
- [40] C. Cui, Y. Ma, X. Cao, W. Ye, and Z. Wang, "Drive as you speak: Enabling human-like interaction with large language models in autonomous vehicles," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2024, pp. 902–909.
- [41] D. Fu, X. Li, L. Wen, M. Dou, P. Cai, B. Shi, and Y. Qiao, "Drive like a human: Rethinking autonomous driving with large language models," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2024, pp. 910–919.
- [42] Y. Wang, R. Jiao, S. S. Zhan, C. Lang, C. Huang, Z. Wang, Z. Yang, and Q. Zhu, "Empowering autonomous driving with large language models: A safety perspective," *arXiv preprint arXiv:2312.00812*, 2023.
- [43] Y.-J. Wang, B. Zhang, J. Chen, and K. Sreenath, "Prompt a robot to walk with large language models," *arXiv preprint arXiv:2309.09969*, 2023.
- [44] L. Wen, D. Fu, X. Li, X. Cai, T. Ma, P. Cai, M. Dou, B. Shi, L. He, and Y. Qiao, "Dilu: A knowledge-driven approach to autonomous driving with large language models," *arXiv preprint arXiv:2309.16292*, 2023.
- [45] S. Sural, R. Rajkumar *et al.*, "Contextvlm: Zero-shot and few-shot context understanding for autonomous driving using vision language models," *arXiv preprint arXiv:2409.00301*, 2024.
- [46] S. Tang, Z. Zhang, J. Zhou, L. Lei, Y. Zhou, and Y. Xue, "Legend: A top-down approach to scenario generation of autonomous driving systems assisted by large language models," in *Proceedings of the 39th IEEE/ACM International Conference on Automated Software Engineering*, 2024, pp. 1497–1508.
- [47] T. Cai, Y. Liu, Z. Zhou, H. Ma, S. Z. Zhao, Z. Wu, and J. Ma, "Driving with regulation: Interpretable decision-making for autonomous vehicles with retrieval-augmented reasoning via llm," *arXiv preprint arXiv:2410.04759*, 2024.
- [48] Z. Xu, Y. Zhang, E. Xie, Z. Zhao, Y. Guo, K.-Y. K. Wong, Z. Li, and H. Zhao, "Drivept4: Interpretable end-to-end autonomous driving via large language model," *IEEE Robotics and Automation Letters*, 2024.
- [49] T. Deruyttere, D. Grujicic, M. B. Blaschko, and M.-F. Moens, "Talk2car: Predicting physical trajectories for natural language commands," *Ieee Access*, vol. 10, pp. 123 809–123 834, 2022.
- [50] T. I. Fossen, "Handbook of marine craft hydrodynamics and motion control," *John Wiley & Sons Ltd*, 2011.
- [51] H. Imazu, "Research on collision avoidance manoeuvre," *Tokyo University of Marine Science and Technology: Tokyo, Japan*, 1987.
- [52] R. Sawada, K. Sato, and T. Majima, "Automatic ship collision avoidance using deep reinforcement learning with lstm in continuous action spaces," *Journal of Marine Science and Technology*, vol. 26, no. 2, pp. 509–524, 2021.
- [53] M. Akdağ, T. A. Pedersen, T. I. Fossen, and T. A. Johansen, "Hierarchical collision avoidance algorithm with route exchange concept for autonomous and conventional ships," *Authorea Preprints*, 2023.